

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Eighth Semester B.E. (Credit System) Degree Examinations

May - June 2023

19MA8X01 – GRAPH THEORY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

- With the help of a graph explain Konigsberg Bridge problem.
 - Define bipartite Graph & complete bipartite graph. Show that if a graph G is bipartite then every cycle in G is of even length.
 - Prove that a tree with two or more vertices contains at least two leaves.
- Define cut point, cycle, loop, subgraph, pendant vertex, bridge, path of a graph.
 - Define isomorphism of two graphs with example.
 - Draw Petersen graph and Kuratowski's second graph.
 - For the following graph find i) any closed walk ii) vertex connectivity iii) edge connectivity.

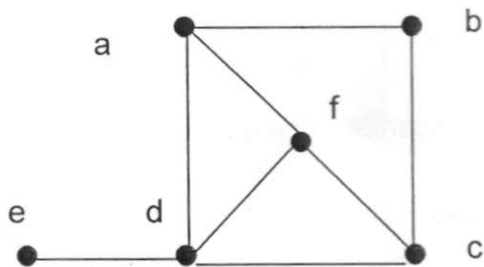
Marks BT* CO* PO*

7 L*1 1 1

7 L1 1 1

6 L1 2 1

7 L1 1 1



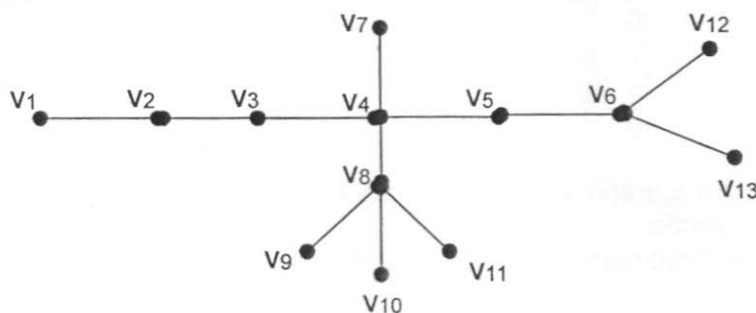
7 L3 2 2

- Suppose that a tree T has two vertices of degree 2, four vertices of degree 3, and three vertices of degree 4. Find the number of pendant vertices in T. ii) Define binary tree. Show that if T is a binary tree with n vertices, then the number of pendant vertices is $\frac{n+1}{2}$.

6 L3 2 2

- If G is a graph with n vertices and k components, prove that the maximum number of edges is $\frac{(n-k)(n-k+1)}{2}$.
 - Find the eccentricities of all the vertices, centre and radius of the graph;

7 L1 1 1



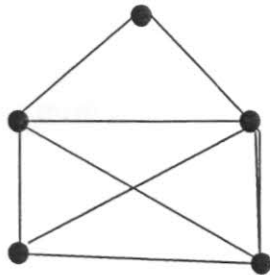
7 L3 2 2

- c) Distinguish between Euler graph and Hamiltonian graph. Give example.

6 L3 2 1

Unit - II

4. a) Find the chromatic polynomial of



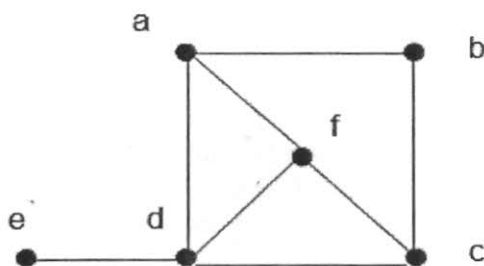
- b) If G is a planar connected graph, prove that $n - m + r = 2$ where n, m, r are the number of vertices, edges and regions respectively.
c) Define incidence matrix of a graph and mention two of its properties. Write the graph G whose incident matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

7 L3 3 2

7 L1 3 1

5. a) State and prove five color theorem.
b) Define path matrix of a graph. Write any two properties. Find a path matrix from a to e for the graph given below.



6 L3 4 2

7 L1 3 1

- c) i) A connected planar graph G has 9 vertices with degrees 2, 2, 3, 3, 3, 4, 5, 6, 6. Find the number of regions of G .
ii) Can a connected simple graph G with $n=1000$ vertices and $m=3000$ edges be planar? Give reason.

7 L3 4 2

6 L3 3 2

6. a) Define Adjacency matrix of graph. Write any two properties. Write an adjacency matrix of K_4 . Find the graph corresponding to the

following adjacency matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

7 L3 4 2

- b) Define chromatic number of a graph. Prove that every tree with two or more vertices is bi-chromatic.
c) Define circuit matrix and cut-set matrix. Write 2 properties.

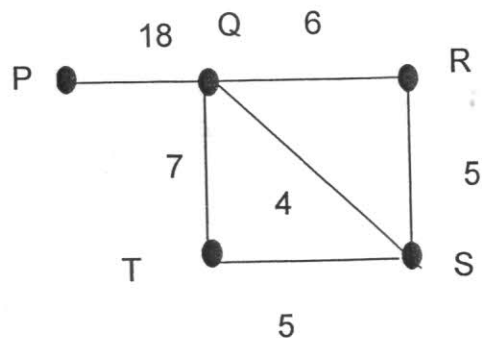
7 L1 3 1

6 L3 4 1

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- b) State Max-flow and min-cut theorem. Find the maximum flow possible between the vertices P and S in the network shown in below figure.



- c) Prove that a graph is connected if and only if it has a spanning tree.

7	L3	5	2
6	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
